Internet traffic forecasting

LSTM, GRU and Transformer models
for intelligent bandwidth prediction

CESNET-TimeSeries24 institution 0
An offline forecasting and capacity-allocation experiment

Research scope

Forecast future demand

Compare three neural models with persistence and a four-observation moving average.

Evaluate capacity decisions

Replay 10-minute forecasts at equal safety margins and measure the tradeoff between shortfalls and spare capacity.

Experimental scope: one institution, one fixed seed, historical replay.

CESNET-TimeSeries24 data

40,298

observations at nominal 10-minute intervals

9 Oct 2023 to 14 Jul 2024

Institution 0, selected after comparing 283 institutions before training.

Recorded demand (Mbps) = bytes × 8 ÷ (600 × 1,000,000)

Recorded bidirectional traffic is a proxy for demand. Physical link utilization needs topology and routing.

Preparation and measurement safeguards

Timestamp continuity	Six continuous blocks. No window crosses a gap or an excluded interval.
Outage policy	Exclude 20 May to 5 June 2024 and guard both ends of each block.
Causal features	Past demand, clock cycles, missingness and trailing mean/standard deviation.
Training statistics	Fit each horizon’s scalers on training windows only.

37,838 eligible observations. Raw measurements remain available for inspection.

Forecasting protocol

Three-hour history. Separate direct forecasts at 10, 30 and 60 minutes.

Horizon	Training	Validation	Test
10 minutes	28,129	4,029	5,554
30 minutes	28,121	4,029	5,548
60 minutes	28,109	4,029	5,539

Chronological 70/15/15 cut points precede exclusions. Target completion timestamps assign the split.

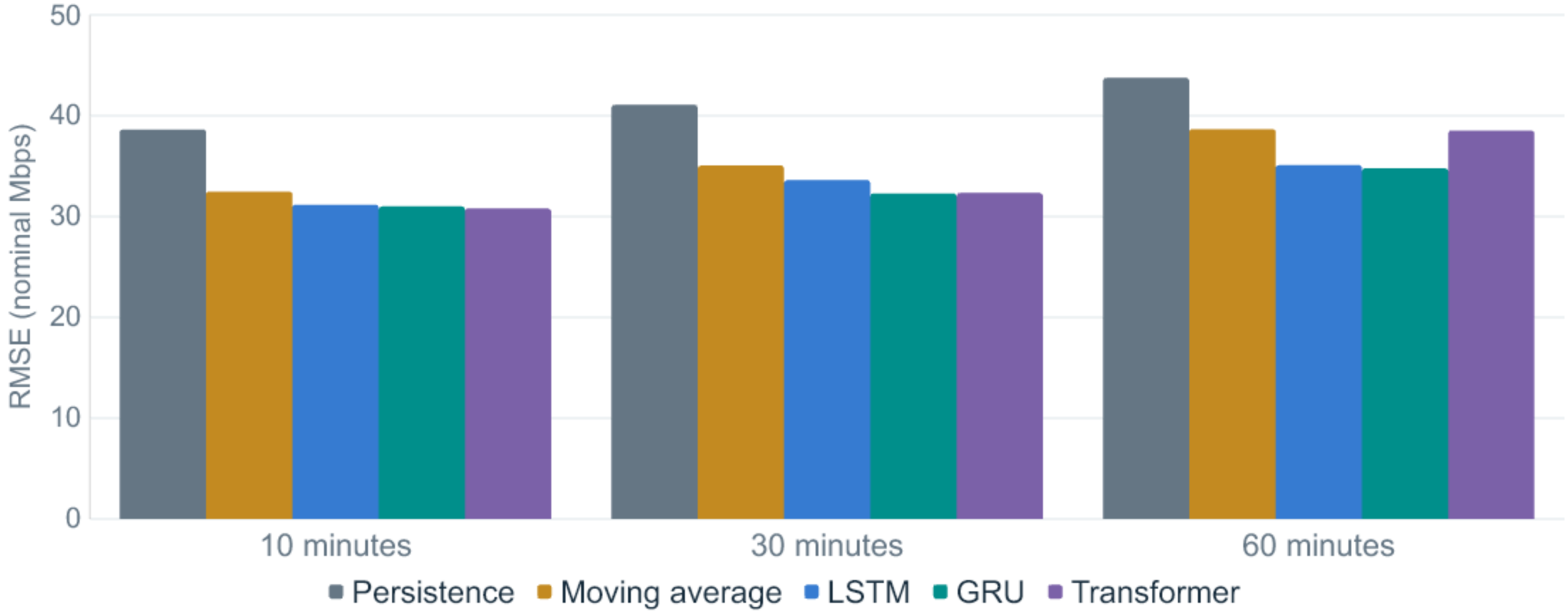
Models and training settings

Predictor	Structure	Parameters
Persistence	Latest observation	0
Moving average	Last four observations	0
LSTM	2 layers, 64 hidden units	52,289
GRU	2 layers, 64 hidden units	39,233
Transformer	2 layers, dimension 64, 4 heads	67,585

Adam 0.001 / Batch 64 / Maximum 30 epochs

Seed 42. Validation early stopping with patience five. Lowest validation MSE selects each checkpoint.

Test forecasting error across horizons



Lower is better. Target counts differ slightly across horizons. One fixed seed.

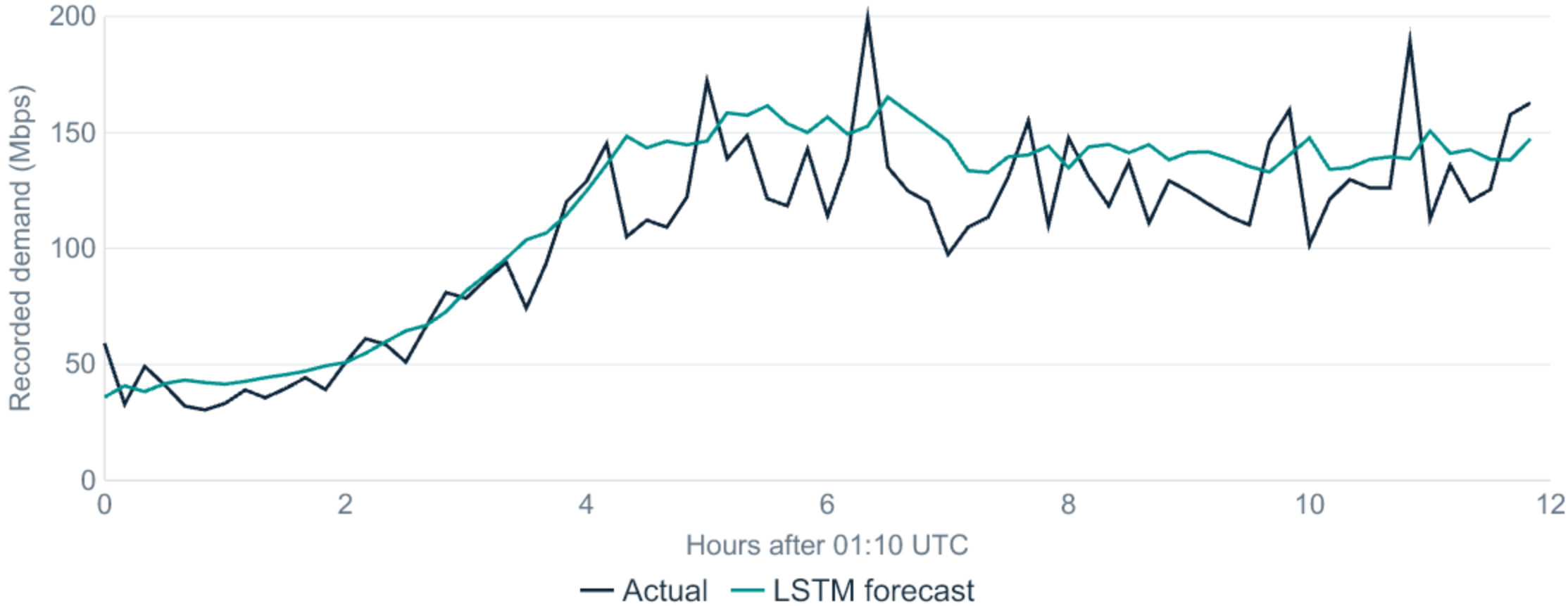
60-minute results

Predictor	MAE (Mbps)	RMSE (Mbps)	MAPE (%)	R^2
Persistence	29.115	43.754	28.98	0.448
Moving average	27.025	38.668	27.97	0.569
LSTM	24.704	35.091	25.32	0.645
GRU	24.506	34.785	25.67	0.651
Transformer	29.235	38.537	32.53	0.572

GRU has the lowest absolute errors. LSTM has the lowest percentage error.

5,539 test targets. All targets are positive. No MAPE exclusions.

Historical forecast replay



First 72 chronological test targets on 2024-06-06. Scores use the full eligible test split.

Capacity allocation policy

Capacity = max(0, forecast) × (1 + margin)

Reactive demand uses the latest observation.
Proactive demand uses a saved model forecast.

Example: forecast 100 Mbps, margin 10%, actual 200 Mbps

110 Mbps capacity. 90 Mbps shortfall. 6,750 MB deficit over 600 seconds.

Deficit MB = max(actual − capacity, 0) × 600 ÷ 8. An abstract capacity pool.

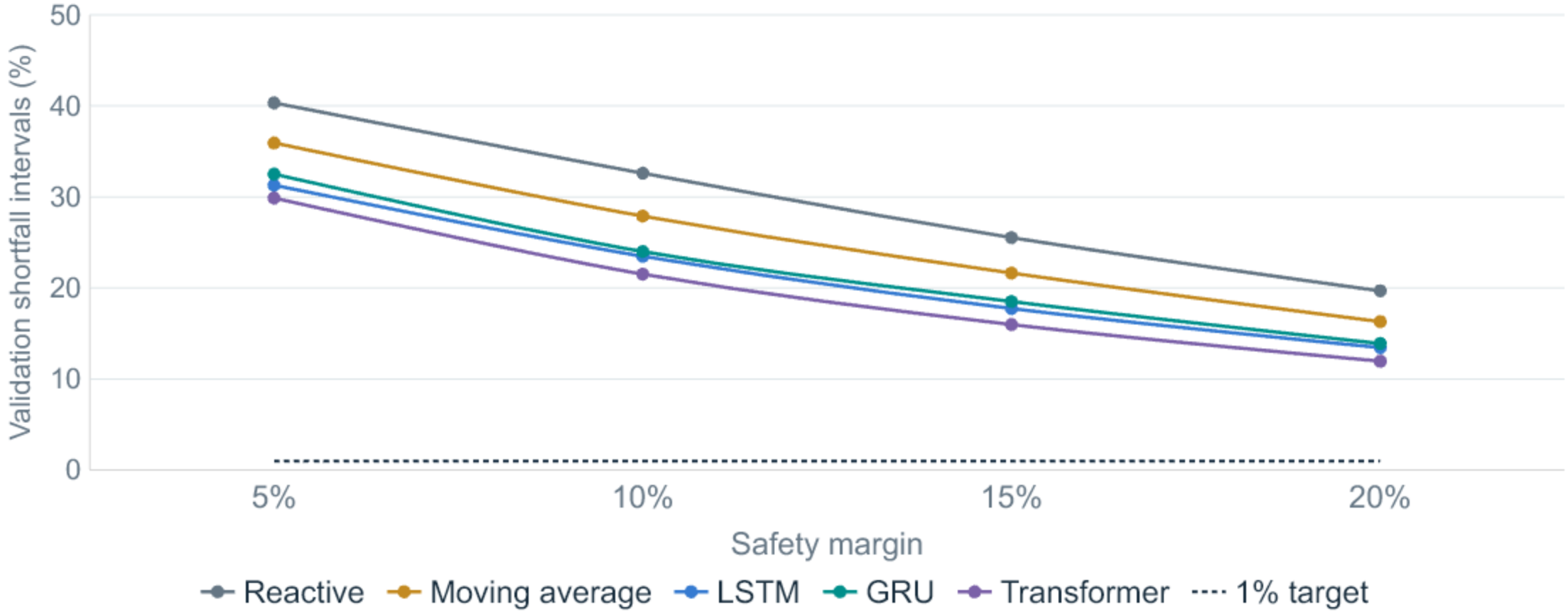
Allocation at an equal 20% margin

Policy	Shortfalls (%)	Capacity (Mbps)	Unused (Mbps)	Deficit (GB)
Reactive	23.44	140.21	28.48	2127.20
Moving average	18.85	140.19	27.39	1685.30
LSTM	10.86	148.67	34.47	1102.08
GRU	11.09	147.98	33.89	1143.21
Transformer	10.97	146.69	32.60	1146.18

Fewer shortfalls can require more capacity and leave more capacity unused.

All policies use 5,554 test targets. Deficit GB is decimal MB ÷ 1,000.

The 1% shortfall target remains unmet



4,029 validation intervals. Every policy uses the 20% fallback. This does not meet the 1% criterion.

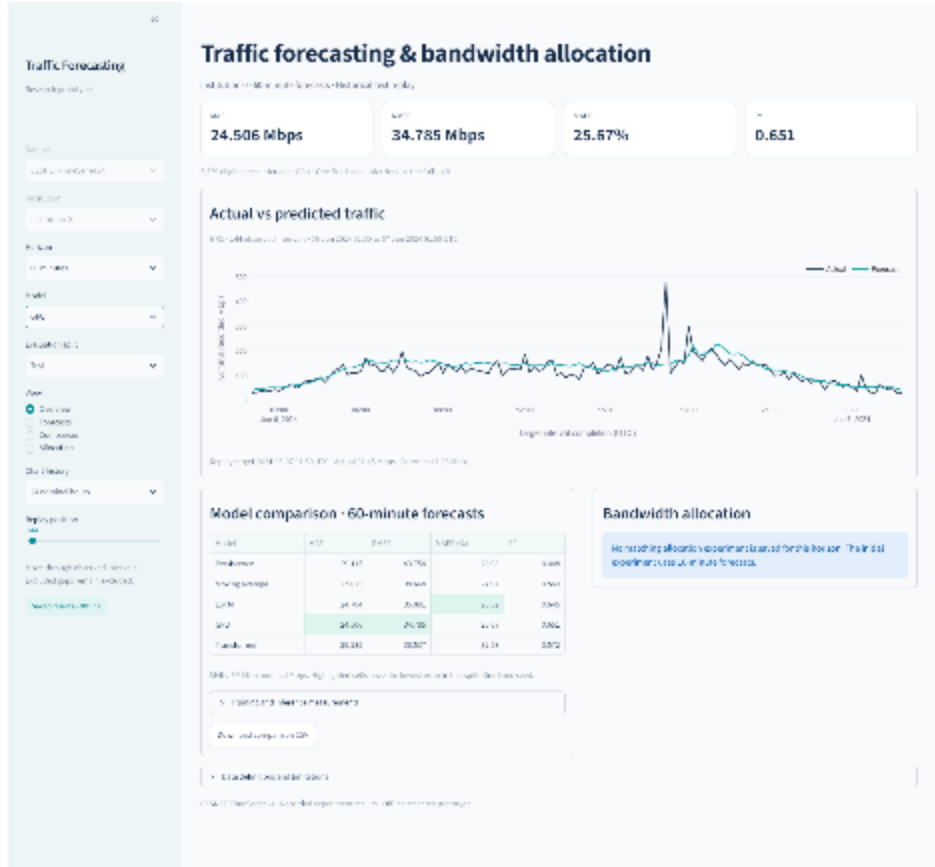
Local dashboard demonstration

Model and horizon

Actual vs predicted

Historical replay

Allocation tradeoffs



Saved results only. Forecasting supports all three horizons. Allocation evaluates the 10-minute horizon.

Findings and limits

Model rankings change with the forecast horizon

Transformer leads at 10 minutes. GRU leads at 30 and 60 minutes by test RMSE.

Capacity savings and service targets need separate evaluation

One institution and one seed limit generalization. The 1% shortfall target remains unmet.

Next research: multiple institutions and seeds, uncertainty estimates, and validation with routing or live collection.

Conclusion

A reproducible prototype for proactive capacity planning

Transformer gives the lowest test RMSE at 10 minutes.
GRU leads at 30 and 60 minutes.

At a 20% margin, LSTM reduces test shortfalls from 23.44% to 10.86%. Mean capacity rises from 140.21 to 148.67 Mbps.

One institution and one seed. The 1% validation target remains unmet. Live deployment needs further validation.